

TASK 2 — PROJECT SUBMISSION REPORT

Human Resources Analytics

HR Workforce & Attrition Dashboard

Interactive Power BI report analysing workforce turnover, satisfaction, performance and recruitment/hiring outlook

1,470 Total employees	16.1% Overall attrition rate	30.5% Attrition rate, overtime staff	43.0% Leavers with ≤2 yrs tenure	311 Projected hires needed (5% growth)
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Deliverable: HR-Workforce-Attrition-Dashboard.pbix | Data source: HR-Employee-Attrition.csv (1,470 employee records)
Tool: Microsoft Power BI Desktop

1. Objective

Create a dashboard to analyse and optimize HR processes and workforce management, using the HR-Employee-Attrition dataset (1,470 employee records) provided for Task 2.

2. Requirements Coverage

The table below maps each stated requirement to the specific page and visual that addresses it.

Requirement	Where addressed
Track recruitment metrics	Page 5 — New Hires, New Hires by Department (proxy metrics based on tenure, since the dataset has no hire-date/applicant fields)
Track employee turnover rates	Pages 1 & 3 — Attrition Rate by Department, Age Group, Overtime, Job Level, Tenure Band, Marital Status
Analyse employee satisfaction and performance data	Page 2 — Job/Environment/Work-Life satisfaction by department, % Rated Top Performer by Job Role, Attrition Rate by Job Satisfaction
Implement predictive analytics for forecasting hiring needs	Page 5 — Key Influencers (Attrition = Yes), What-If growth parameter, Projected Hiring Need measure
Deliverable: interactive Power BI report aiding HR decision-making	All pages — slicers, reset bookmarks, drill-through, What-If slider, written insight panels

3. Dashboard Structure

The report is built as five interconnected pages, each with slicers and a reset-filter control for interactivity.

Page	Title	Purpose
1	Workforce Analysis	Headcount, department and job-role mix, attrition rate by department and by age band
2	Employee Satisfaction & Performance	Job/environment/work-life satisfaction by department, top-performer rate by job role, attrition rate by satisfaction level
3	Workforce Planning & Attrition Risk	Attrition rate by overtime, job level, tenure band and marital status
4	Employee Demographics & Workforce Profile	Gender, marital status, education field and education level breakdowns
5	Recruitment & Hiring Forecast	Recruitment proxy metrics, Key Influencers predictive analysis, What-If hiring-need forecast

4. Key Findings

4.1 Turnover

- Overall attrition rate is 16.1% (237 of 1,470 employees).
- Employees working overtime leave at 30.5%, nearly three times the rate of those who don't (10.4%).
- Attrition is highest among employees under 25 (39.2%) and falls steadily with age, before rising slightly again for the 55-64 group (15.9%).
- 43% of all leavers had two years or less of tenure — early-career retention is the single largest lever available to reduce turnover.
- By department, Sales has the highest attrition rate (20.6%), followed by HR (19.0%) and R&D (13.8%).
- Single employees leave at more than double the rate of married employees (25.5% vs 12.5%).

4.2 Satisfaction & Performance

- Attrition falls as job satisfaction rises: 22.8% at the lowest satisfaction level down to 11.3% at the highest.
- Average satisfaction scores (job, environment, work-life) are broadly similar across departments (2.6–2.9 on a 1–4 scale), so department alone is a weak predictor — individual satisfaction level is the stronger signal.
- Managers have the highest share of top-rated performance reviews (19.6%); Research Directors have the lowest (10.0%).

4.3 Recruitment & Forecasting

- The dataset contains no hire-date or applicant data, so recruitment activity is estimated using tenure as a proxy: 215 employees (15% of the workforce) have one year or less of service, concentrated in Research & Development (149) and Sales (61).
- A Key Influencers analysis targeting Attrition = Yes confirms the strongest individual drivers: age 21 or under (3.57x more likely to leave), working overtime (2.93x), one year or less of tenure (2.70x), and monthly income under 2,800 (2.60x).
- At the current attrition rate and a default 5% headcount growth assumption, the model projects a need for approximately 311 new hires — 237 to replace expected leavers and 74 to support growth. This figure updates live as the growth assumption is adjusted.

5. Methodology

5.1 Data Model

The single-table dataset (HR-Employee-Attrition) was extended with calculated columns and measures to support banding, sorting, and rate-based (rather than count-based) visuals, since raw counts alone were misleading for comparing groups of very different sizes.

Field	Type	Purpose
Age Band / Age Sort	Column	Groups Age into readable bands (Under 25 ... 55-64) with a numeric sort order
Tenure Band / Tenure Sort	Column	Groups YearsAtCompany into 0-2, 3-5, 6-10, 10+ bands
Education Level	Column	Maps Education codes 1-5 to Below College ... Doctor

Field	Type	Purpose
Attrition Rate	Measure	Attrition count ÷ total headcount, used throughout instead of raw counts
Rated 4 Pct	Measure	Share of employees with the top Performance Rating (4), since ratings only take values 3 and 4
New Hires	Measure	Employees with ≤ 1 year of tenure — recruitment proxy
Early Attrition Pct	Measure	Share of leavers with ≤ 2 years of tenure
Replacement Hires / Growth Hires / Projected Hiring Need	Measure	Forecast hiring need = current attrition rate $\times$ headcount, plus a growth component set by a What-If parameter

5.2 Predictive Analytics Approach

Two complementary techniques were used to satisfy the forecasting requirement:

- **Key Influencers visual** — an out-of-the-box Power BI AI visual that ranks the factors most associated with an employee leaving (Attrition = Yes), based on the full set of demographic, compensation and engagement fields in the dataset.
- **What-If parameter** — a numeric growth-rate parameter (0%–20%) drives a hiring-need formula: Projected Hiring Need = Replacement Hires (current attrition rate $\times$ headcount) + Growth Hires (headcount $\times$ assumed growth rate). This lets HR test different growth scenarios interactively rather than relying on a single static number.

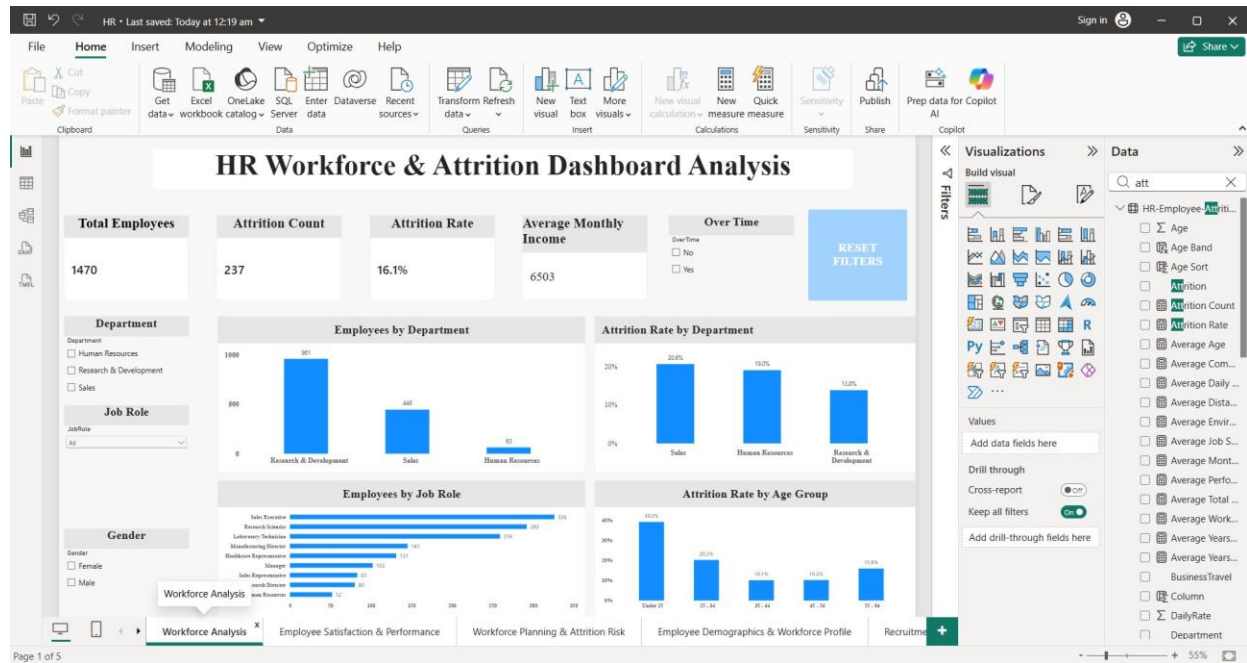
6. Assumptions & Limitations

- The dataset has no hire-date, applicant, or requisition fields, so true recruitment metrics (time-to-hire, source of hire, cost per hire) could not be calculated. "New hires" is a proxy defined as employees with one year or less of tenure.
- Hiring-need forecasting treats the current snapshot attrition rate (16.1%) as if it were an annual rate; a longer historical trend line was not available in this dataset.
- performance Rating in the source data only contains the values 3 and 4, so "share of employees rated 4 (top performer)" was used instead of an average rating, which would otherwise compress into a narrow, uninformative range.
- All satisfaction, environment, and work-life balance scores are on a 1–4 scale as provided by the source data.

7. Conclusion

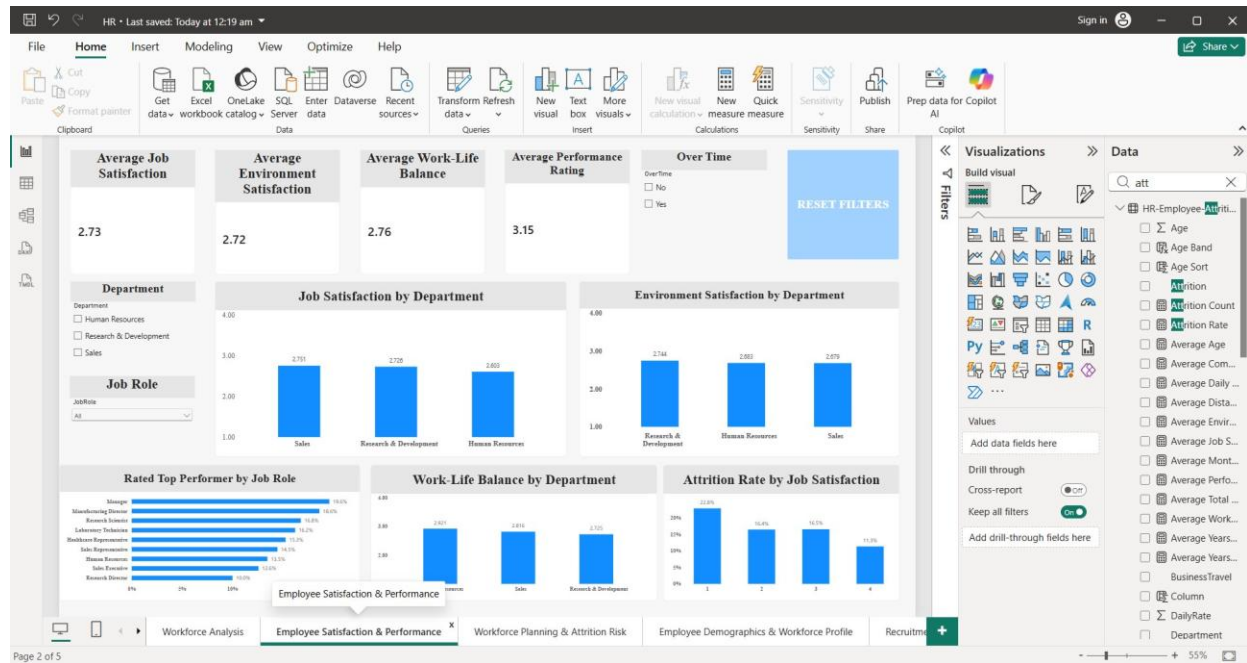
The dashboard satisfies all four key requirements set out in Task 2: recruitment metrics (via tenure-based proxies), turnover rate tracking (department, age, tenure, overtime, marital status), satisfaction and performance analysis (with a direct link to attrition), and predictive analytics for hiring needs (Key Influencers plus a What-If forecast model). The report is fully interactive — slicers, reset buttons, and a growth-rate slider let HR stakeholders explore the data and test scenarios directly, supporting the stated deliverable of "an interactive Power BI report aiding HR decision-making."

Appendix A — Page 1: Workforce Analysis



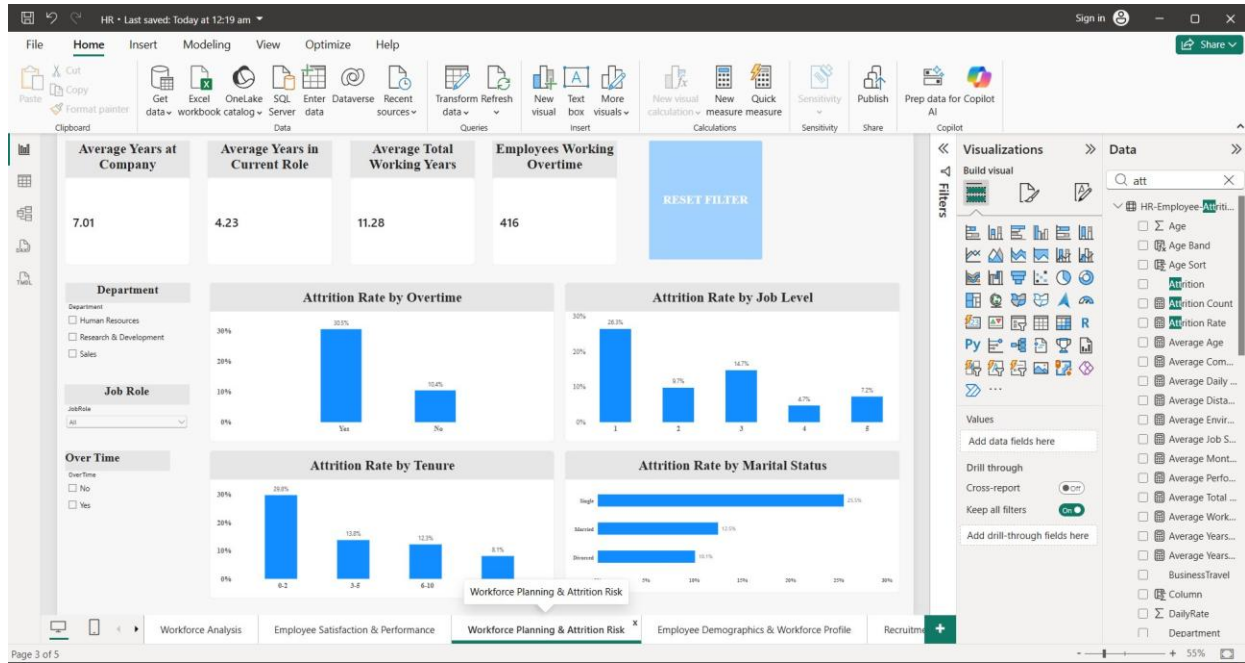
Headcount, department mix, job roles, and attrition rate by department and age band.

Appendix B — Page 2: Employee Satisfaction & Performance



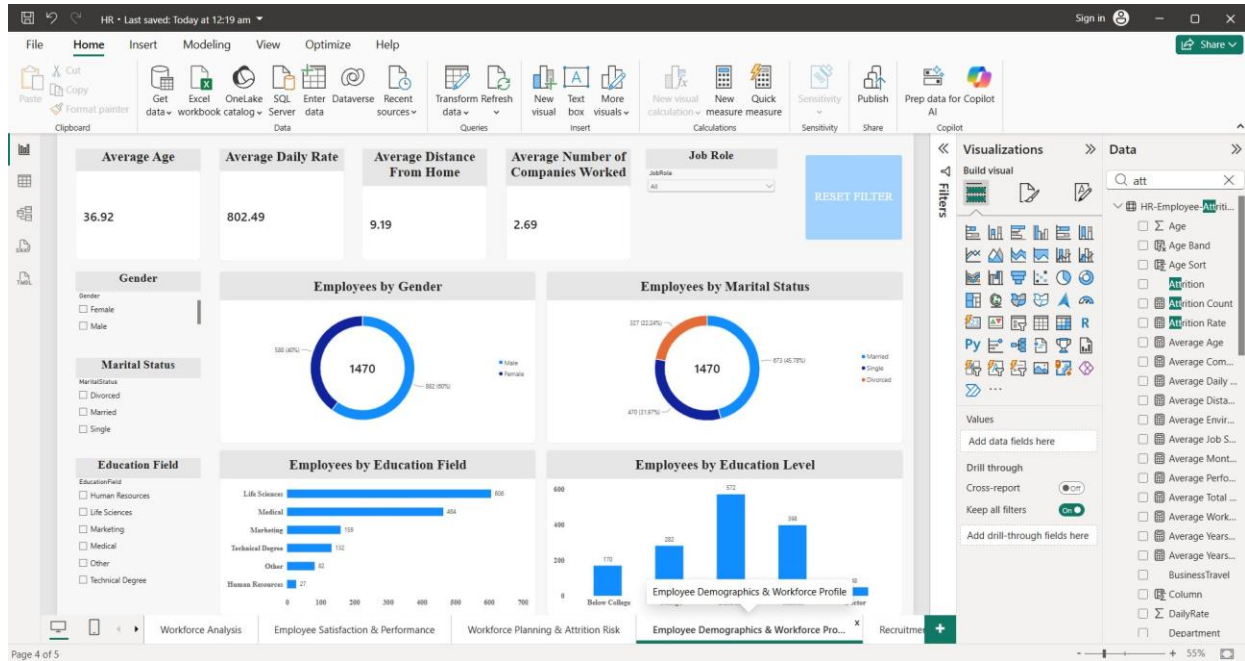
Satisfaction scores by department, top-performer rate by job role, and attrition rate by job satisfaction level.

Appendix C — Page 3: Workforce Planning & Attrition Risk



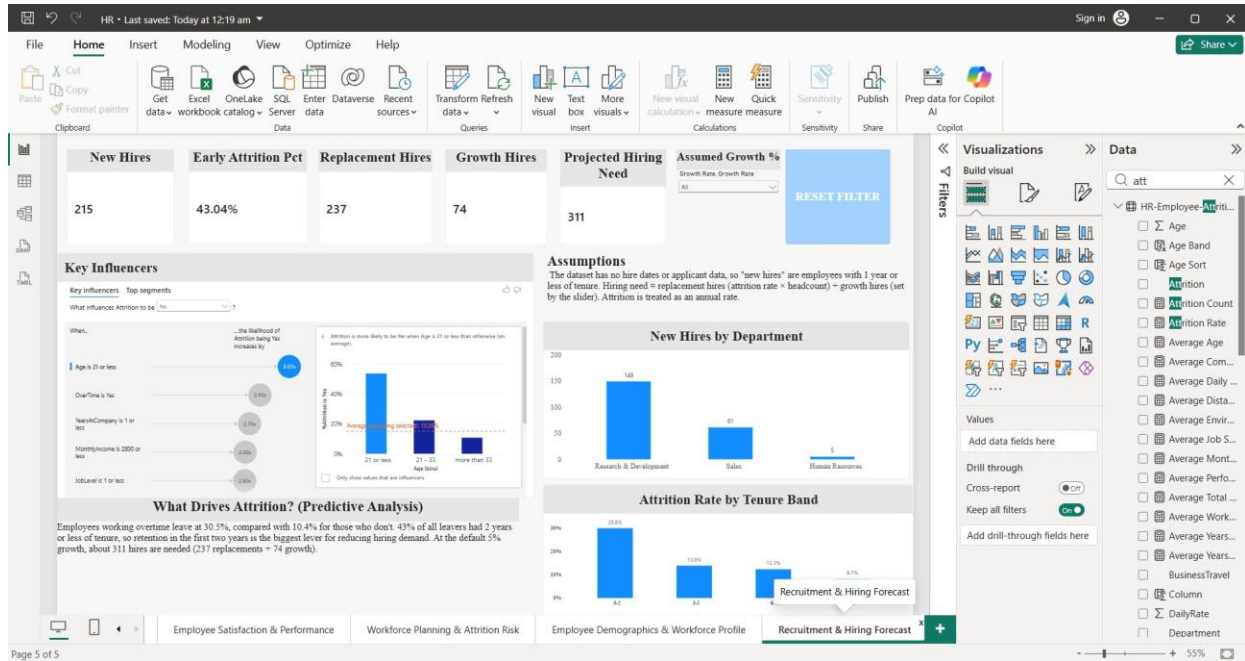
Attrition rate by overtime, job level, tenure band and marital status.

Appendix D — Page 4: Employee Demographics & Workforce Profile



Gender, marital status, education field and education level breakdowns.

Appendix E — Page 5: Recruitment & Hiring Forecast



Recruitment proxy KPIs, Key Influencers predictive analysis, and the What-If hiring-need forecast.